

ECE 6552 – Lecture 23

CONTROL BARRIER FUNCTIONS

April 9 2026

Overview:

- Extend barrier functions to Control Barrier Functions

Additional Reading:

- A. Ames, S. Coogan, M. Egerstedt, G. Notomista, K. Sreenath, and P. Tabuada, “Control Barrier Functions: Theory and Applications,” IEEE Transactions on Automatic Control, 2019.

Control Barrier Functions

Consider a control-affine system

$$\dot{x} = f(x) + g(x)u \quad (1)$$

and a given set $\mathcal{C} = \{x \mid h(x) \geq 0\}$. How can we choose a controller $u(x)$ such that $\mathcal{C}$ is positively invariant?

Recall Barrier Functions:

$$\dot{h}(x) = \nabla h(x)^T f(x) \geq -\alpha(h(x)) \quad \text{for all } x \in \mathbb{R}^n \quad (2)$$

Recall that we also have the following theorem for barrier functions:

Theorem: Barrier Function. If h is a barrier function, then $\mathcal{C} = \{x : h(x) \geq 0\}$ is positively invariant.

Definition: Control Barrier Function. A function h with $\mathcal{C} = \{x \mid h(x) \geq 0\}$ is a control barrier function (CBF) for a control-affine system $\dot{x} = f(x) + g(x)u$ if there exists a locally Lipschitz function $\alpha : \mathbb{R} \rightarrow \mathbb{R}$ satisfying $\alpha(0) = 0$ such that

$$\sup_{u \in \mathbb{R}^m} \nabla h(x)^T (f(x) + g(x)u) \geq -\alpha(h(x)) \quad \text{for all } x \in \mathbb{R}^n. \quad (3)$$

We can also write (3) using Lie derivative notation:

$$\sup_{u \in \mathbb{R}^m} L_f h(x) + L_g h(x)u \geq -\alpha(h(x)) \quad (4)$$

Define

$$U(x) = \{u \in \mathbb{R}^m \mid \nabla h(x)^T (f(x) + g(x)u) \geq -\alpha(h(x))\}. \quad (5)$$

The supremum is the smallest number that is greater than or equal to every element in the set. The supremum must be a real number (cannot be infinity).

Theorem: Invariance from CBF. If h is a control barrier function for (1), then the following hold:

1. $U(x) \neq \emptyset$ for all x ;
2. Any Lipschitz feedback control $u : \mathbb{R}^n \rightarrow \mathbb{R}^m$ satisfying $u(x) \in U(x)$ renders $\mathcal{C}$ invariant;
3. A feedback control is given by

$$u^*(x) = \begin{cases} 0 & \text{if } \nabla h(x)^T f(x) + \alpha(h(x)) \geq 0 \\ \frac{-\nabla h(x)^T f(x) - \alpha(h(x))}{\|\nabla h(x)^T g(x)\|_2^2} (g(x)^T \nabla h(x)) & \text{otherwise.} \end{cases} \quad (6)$$

this is the same thing as writing:

$$u^*(x) = \begin{cases} 0 & \text{if } L_f h(x) + \alpha(h(x)) \geq 0 \\ \frac{-(L_f h(x) + \alpha(h(x))) L_g h(x)^T}{L_g h(x) L_g h(x)^T} & \text{otherwise} \end{cases} \quad (7)$$

A sufficient condition for $u^*(x)$ to be Lipschitz on some domain is that $\nabla h(x)^T g(x) \neq 0$ everywhere on the domain.

Proof. The proof of all three parts is as follows:

1. If $\sup_{u \in \mathbb{R}^m} \nabla h(x)^T (f(x) + g(x)u) < \infty$, then the sup is attained for some u .
2. h becomes a (regular) barrier function for $\tilde{f}(x) = f(x) + g(x)u(x)$ and theorem from previous lecture applies.
3. (Sketch) First, note that $u^*(x)$ is well-defined since $\nabla h(x)^T g(x) \neq 0$ whenever $\nabla h(x)^T f(x) + \alpha(h(x)) < 0$ by CBF condition. $u^*(x)$ can be considered as a composition of 3 Lipschitz functions and is therefore Lipschitz. Finally, we can verify that

$$\nabla h(x)^T (f(x) + g(x)u^*(x)) + \alpha(h(x)) \quad (8)$$

$$= \begin{cases} \nabla h(x)^T f(x) + \alpha(h(x)) & \text{if } \nabla h(x)^T f(x) + \alpha(h(x)) \geq 0 \\ 0 & \text{otherwise} \end{cases} \quad (9)$$

$$\geq 0. \quad (10)$$

Minimum Effort Control

From the above proof, specifically, the condition

$$\nabla h(x)^T (f(x) + g(x)u^*(x)) + \alpha(h(x)) \quad (11)$$

$$= \begin{cases} \nabla h(x)^T f(x) + \alpha(h(x)) & \text{if } \nabla h(x)^T f(x) + \alpha(h(x)) \geq 0 \\ 0 & \text{otherwise,} \end{cases} \quad (12)$$

we conclude that $u^*(x)$ is the “minimum effort” controller, *i.e.*,
 $u^*(x) = \operatorname{argmin}_{u \in U(x)} \|u\|_2^2$.

Example (Cart-Pole System):

Recall the model of the cart-pole system from Lecture 16 (take $m = M = \ell = 1$):

$$\begin{aligned}\ddot{y} = \dot{v} &= \frac{1}{1 + \sin^2 \theta} \left(u + \dot{\theta}^2 \sin \theta - g \sin \theta \cos \theta \right) \\ \ddot{\theta} &= \frac{1}{1 + \sin^2 \theta} \left(-u \cos \theta - \dot{\theta}^2 \cos \theta \sin \theta + 2g \sin \theta \right)\end{aligned}\quad (13)$$

where $v = \dot{y}$ is velocity. Take as the state $x = [y \ v \ \theta]^T$. Suppose we want v to satisfy

$$-L \leq v \leq L.$$

Choose

$$h(x) = \frac{1}{2}(-v^2 + L^2) \quad (14)$$

$$\alpha(s) = \gamma s, \quad \gamma > 0. \quad (15)$$

Then

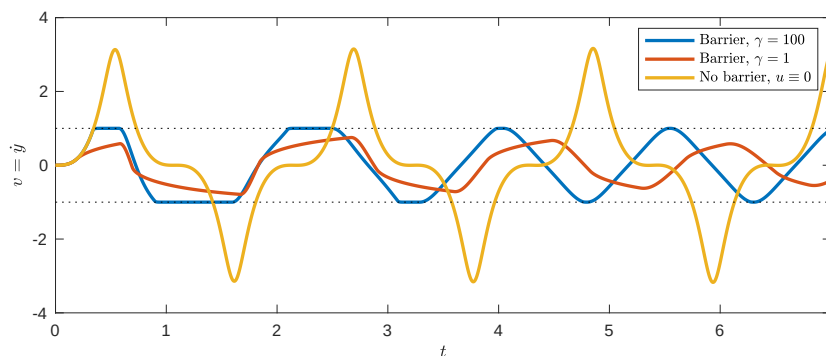
$$\nabla h(x)^T f(x) = L_f h(x) = \frac{-v}{1 + \sin^2(\theta)} (\dot{\theta}^2 \sin \theta - g \sin \theta \cos \theta) \quad (16)$$

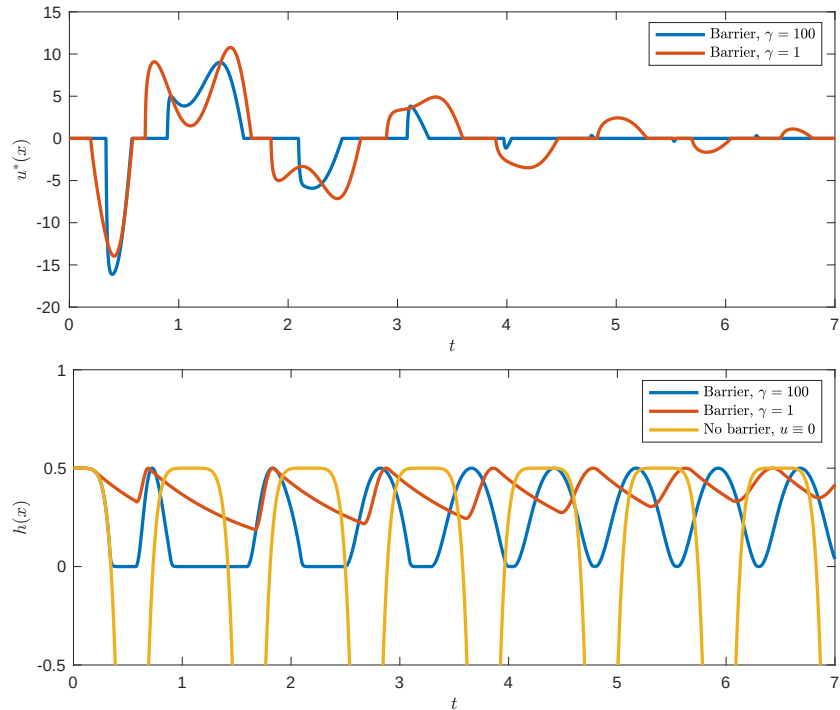
$$\nabla h(x)^T g(x) = L_g h(x) = \frac{-v}{1 + \sin^2(\theta)} \quad (17)$$

$$\alpha(h(x)) = \gamma h(x) \quad (18)$$

and $u^*(x)$ constructed as above.

The figures below show results for $x_0 = [y_0 \ v_0 \ \theta_0 \ \dot{\theta}_0]^T = [0 \ 0 \ \pi/2 \ 0]^T$ using the $u^*(x)$ from the theorem.





Controller Synthesis as Optimization Problem

For fixed x , the CBF constraint is *affine* in u ! Then we can define a *convex* program to compute a control input at each time instant:

$$\begin{aligned} u(x) = \arg \min_{\mu} \quad & C(\mu, x) \\ \text{subject to} \quad & \nabla h(x)^T f(x) + \nabla h(x)^T g(x) \mu \geq -\alpha(h(x)) \end{aligned} \quad (19)$$

where $C(\mu, x)$ is some cost function that is convex in μ for each fixed state x .

Example 1: Suppose $k(x) : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is some nominal feedback controller designed for some other purpose (e.g., performance objectives). Can choose $C(\mu, x) = \|\mu - k(x)\|_2^2$. The result is a quadratic program (with affine constraints) to compute $u(x)$ at each x .

- Raises questions about solving a QP in real-time online, care must be taken with discretization values, etc.
- Convex solvers are fast enough that they can be included “in-the-loop” and have been for applications like stable bipedal locomotion, quadrotor control

Example 2: Consider our controlled pendulum:

$$\dot{x} = \frac{d}{dt} \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \dot{\theta} \\ -\frac{g}{l} \sin(\theta) + u \end{bmatrix} \implies \dot{x} = \begin{bmatrix} x_2 \\ -\frac{g}{l} \sin(x_1) + u \end{bmatrix}$$

Let's assume that we want to limit the velocity of the pendulum, we can define our safe set as:

$$\mathcal{C} = \{x \mid h(x) := v_{\max}^2 - x_2^2 \geq 0\}$$

Taking the derivative of $h(x)$ we have:

$$\begin{aligned} \dot{h} &= -2x_2\dot{x}_2 = -2x_2\left(-\frac{g}{l}\sin(x_1) + u\right) \\ &= \underbrace{2x_2\frac{g}{l}\sin(x_1)}_{L_f h} \underbrace{-2x_2 u}_{L_g h} \end{aligned}$$

Thus, safety can be enforced via the CBF condition:

$$L_f h + L_g h u \geq -\alpha(h)$$

where $\alpha(h) = \gamma h$ for some $\gamma > 0$. This can be enforced along with a tracking controller on the pendulum via the aforementioned QP formulation¹:

$$\begin{aligned} u^* &= \underset{u}{\text{minimize}} \quad \|u - u_{des}\|_2^2 \\ &\text{subject to } L_f h + L_g h u + \gamma h \geq 0 \end{aligned}$$

¹ An implementation of this example is provided [online](#).

Note: For CBFs to be implemented in this way, we need to ensure that $\nabla h(x)^T g(x) = L_g h(x) \neq 0$ for all x in the domain of interest. However, this would preclude us from selecting a CBF to limit the pendulum position, i.e., $h(x) := \theta_{\max}^2 - x_1^2$. In this case, we need higher-order CBFs. We will cover these in the next lecture.